Ex 8: Interactive Dashboards with Tableau

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
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Ex8: Interactive Dashboards with Tableau

Learning Objective:
The objective of this experiment is to enable students to understand and implement interactive data visualisation and dashboard design using Tableau.

Problem Statement:
Create an Interactive Dashboard in Tableau using the Sample – Superstore dataset from kaggle that includes:

- Bar chart (Sales by Category or Region)
- Line chart (Sales trend over time)
- Scatter plot (Profit vs Sales by Sub-Category)
- Interactive filters and dashboard actions

Tasks to Perform:

1. Open Tableau and Load dataset.
2. Explore the Data
3. **Create Worksheet 1 – Bar Chart (Sales by Category)**
   a. Click the **Sheet 1** tab → rename it `Bar_Sales_by_Category`.
   b. Drag `Category` to **Columns**.
   c. Drag `SUM(Sales)` to **Rows**.
   d. Sort the bars in descending order.
   e. Drag `Region` to **Color** → this gives different colors for each region.
   f. Add **Labels**: Click **Label** → check "Show Mark Labels".
   g. (Optional) Add filter: Drag `Region` to **Filters** → select all → then right-click it → "Show Filter".
4. **Create Worksheet 2 – Line Chart (Sales Trend Over Time)**
   a. Click **New Worksheet** → rename it `Line_Sales_Trend`.
   b. Drag `Order Date` to **Columns**.
   c. Right-click `Order Date` → select **Month** (continuous).
   d. Drag `SUM(Sales)` to **Rows**.
   e. Drag `Region` to **Color** (to show separate lines per region).
   f. Format the axis (if needed) → Right-click axis → "Format" → Currency.
   g. Show filter for Region or Category (optional).
5. **Create Worksheet 3 – Scatter Plot (Profit vs Sales)**
   a. Click **New Worksheet** → rename it `Scatter_Profit_vs_Sales`.
   b. Drag `SUM(Sales)` to **Columns**.
   c. Drag `SUM(Profit)` to **Rows**.
   d. Drag `Sub-Category` to **Detail** (each point = one Sub-Category).
   e. Drag `Category` to **Color**.

f. Optionally:
   i. Drag `Segment` to **Shape** (to add another dimension).
   ii. Add **Reference Line**: Analytics Pane → drag "Average Line" to both axes → "Table".

6. **Build the Dashboard**
   a. Click **New Dashboard** (bottom tab) → rename it `Sales Dashboard`.
   b. On the left, in the **Sheets** pane, drag and drop your 3 worksheets:
      i. Line chart at top
      ii. Bar chart bottom-left
      iii. Scatter plot bottom-right
   c. Adjust size (e.g., 1200×800).
   d. Use **containers** (horizontal/vertical) for cleaner alignment.
   e. Add **Title** → "Sales and Profit Dashboard".
7. **Add Interactivity**

(a) Add Filters

1. In Dashboard view → click dropdown on top-right of each sheet → "Show Filter".
2. Show filters for fields like **Region**, **Category**, or **Year**.

(b) Add Actions

1. Menu → **Dashboard → Actions → Add Action → Filter…**
2. Example:
   - **Source Sheet**: Bar chart (Sales by Category)
   - **Target Sheets**: Line and Scatter plots
   - **Run action on**: Select
   - **Clearing the selection**: Show all values
3. Add another **Highlight Action** (optional):
   - So hovering over a Sub-Category highlights the same category in other charts.

8. **Publish or Export**
   a. **To Tableau Public**: *File → Save to Tableau Public → sign in → publish.*
   b. **To PDF/Image**: *File → Export → Image/PDF.*

Aim

To design an interactive Tableau dashboard on the Sample – Superstore dataset containing a bar chart, a line chart, a scatter plot, interactive filters and dashboard actions.

Procedure

1. Open Tableau and connect to the Sample – Superstore data source; explore the fields in the Data pane.
2. Worksheet 1 (Bar_Sales_by_Category): put Category on Columns and SUM(Sales) on Rows, sort descending, drag Region to Color, show mark labels and optionally show the Region filter.
3. Worksheet 2 (Line_Sales_Trend): put Order Date (Month, continuous) on Columns and SUM(Sales) on Rows, drag Region to Color, format the axis as Currency.
4. Worksheet 3 (Scatter_Profit_vs_Sales): put SUM(Sales) on Columns and SUM(Profit) on Rows, drag Sub-Category to Detail and Category to Color; optionally add Segment to Shape and average reference lines.
5. Create a new dashboard named Sales Dashboard: line chart at top, bar chart bottom-left, scatter plot bottom-right; set the size to 1200 × 800 and use containers for alignment; add the title “Sales and Profit Dashboard”.
6. Add interactivity: show filters for Region, Category and Year; add a Filter action (source: bar chart; targets: line and scatter) and an optional Highlight action.
7. Publish to Tableau Public or export the dashboard as an image/PDF.

Implementation

Tableau is a drag-and-drop tool, so no program code is required. The worksheet configuration used is summarised below.

Worksheet	Columns	Rows	Marks
Bar_Sales_by_Category	Category	SUM(Sales)	Color: Region; Labels on; sorted descending
Line_Sales_Trend	MONTH(Order Date) continuous	SUM(Sales)	Color: Region; axis format: Currency
Scatter_Profit_vs_Sales	SUM(Sales)	SUM(Profit)	Detail: Sub-Category; Color: Category; Shape: Segment

Result

An interactive Sales and Profit Dashboard was created in Tableau with a bar chart, line chart and scatter plot. Region, Category and Year filters and a filter action from the bar chart to the other sheets allow the user to drill into the data interactively.